

2-Mark Questions & Answers -- DBMS Module 5: Query Optimization & Transaction Concurrency Control

Q1. What is query optimization

Query Optimization is the DBMS process of selecting the most efficient physical evaluation plan for a given SQL query to minimize Disk I/O and CPU execution time.

Q2. Define transaction and ACID properties.

A Transaction is a logical unit of database work.

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Q3. What is conflict serializability

A schedule is Conflict Serializable if it is conflict equivalent to some serial schedule (it can be transformed into a serial schedule by swapping non-conflicting operations).

Q4. Define Shared and Exclusive locks.

* Shared Lock (S): Allows multiple transactions to read a data item concurrently.

* Ex

Q5. What is the phantom phenomenon

The Phantom Phenomenon occurs when a concurrent transaction inserts or deletes tuples satisfying a search condition while another transaction is performing range queries over that condition.